Ricardo (Zhicong) Xie

Systems / Project Engineering Intern | Requirements, Risk and Technical Coordination

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LinkedIn: https://www.linkedin.com/in/ricardo-xie-6394643b4 | Portfolio: https://zxie588-alt.github.io | GitHub: https://github.com/zxie588-alt

NZ student visa | Full-time eligible over scheduled summer break 17 Nov 2026 - late Feb 2027 | No employer visa support needed for this internship | Open NZ-wide

PROFILE

- Mechanical master's student with team-lead evidence in requirements, trade studies, risk and verification.
- Best fit: project engineering, systems engineering, infrastructure and technical coordination internships.
- Project screenshots and source links: https://zxie588-alt.github.io

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Study focus and portfolio work in mechanical, building-services, manufacturing and analysis contexts.

Study record/focus: Completed: Engineering Methods, Computational Fluid Dynamics, Composite Materials, Aerospace Systems Engineering. Registered S2: Research Project, Smart Manufacturing, Manufacturing and Industrial Systems.

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Engineering team leadership: led task scoping, workload split and final technical presentation in team design work.
- Workshop practice: prepared AutoCAD/DXF laser-cut geometry and checked cut parts against drawings.
- Hands-on exposure: CNC machining, workholding, cutting fluid, chip formation and 3D printing workflow.
- Lab practice: interpreted force-displacement tests and controlled a robotic arm in training sessions.
- Technical communication: former student news lead; strong project records, photography and presentation visuals.

PROJECT WORK

Team Systems Engineering Feasibility Study

Stakeholders, requirements, ConOps, trade studies, risk, verification

- Led systems-engineering structure for a 5-person technical feasibility study.
- Built stakeholder map, requirements hierarchy, ConOps, trade criteria and risk register.
- Used verification gates to keep orbit, power, thermal, communications and cost assumptions aligned.

Small Residential MVHR / ERV Revit MEP Coordination

Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes

- Built a small NZ residential MVHR/ERV coordination package with Revit MEP, SolidWorks and AutoCAD/DXF evidence.
- Created plan, 3D view, airflow schedule, sheet, review comments and simple clash/coordination notes.
- Documented preliminary airflow, pressure-loss, balancing and commissioning actions for the next design iteration.

Ruggedized Outdoor IoT Sensor Module Design Package

SolidWorks, enclosure design, DFM notes, Keil C, testing workflow

- Designed a SolidWorks enclosure with gasket stack-up, screw bosses, PCB envelope and service access.
- Prepared DFM notes for CNC review, 3D printing and assembly tolerance checks.
- Reviewed load paths around corner, strap-lug and screw-boss regions; built Keil C module structure.

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared AutoCAD/DXF geometry for laser cutting and checked the cut plate against drawing intent.
- Practised CNC workflow: workholding, tool motion, cutting fluid, chip formation and shop safety.
- Reviewed machined gears and shafts for tolerance, edge quality and tool-access constraints.

TECHNICAL SKILLS

- Systems: stakeholder mapping, requirements hierarchy, ConOps, trade studies and risk register.
- Coordination: task alignment, assumption tracking, decision logs and technical reviews.
- Engineering tools: Revit MEP, SolidWorks, AutoCAD/DXF, Abaqus, ANSYS Fluent/CFX and STM32 C.